

Table 2

	Component's name	Purpose	Location
1	Label	To display a label	Palette-->User interface-->Label
2	Button	To trigger events	Palette-->User interface-->Button
3	Text box	To take user input	Palette-->User interface-->Text
4	Horizontal arrangement	To arrange the child components	Palette-->Layout-->Horizontal arrangement
5	Notifier	To display on screen information	Palette-->User interface-->Notifier
6	TinyDB	To store data locally to phone	Palette-->Storage-->TinyDB

purpose of the second screen is to add the students' names so that their attendance can be verified.

1. Drag and drop the components mentioned in Table 2 to the viewer.
2. If you have dragged and placed everything, the layout will look something like what's shown in Figure 8.

We need to configure the Firebase component to connect to the demo project we created using the steps mentioned above. For that, just assign the URL of your project to the Firebase URL property of the Firebase component in the designer.

Now we will head towards the block editor to define the various types of behaviour; so let's discuss the actual functionality that we expect from our application.

1. From the very first screen, the user should be able to scan the barcode.
2. The barcode will consist of the student's name and roll number, separated by a comma.
3. If the student's name from the barcode doesn't match with the database, the user needs to add that particular student by going to the second screen.
4. Once the barcode is scanned, the result will be stored into the Firebase DB with the date on which this task was performed.
5. The project bucket will be named after the month, so that each month, a new folder will be created and attendance records will be separated for each date.

So let's move on and add these types of behaviour using the block

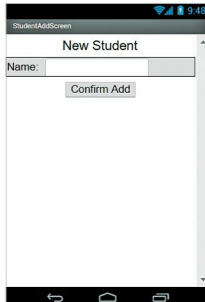


Figure 8: Designer screen for the second screen

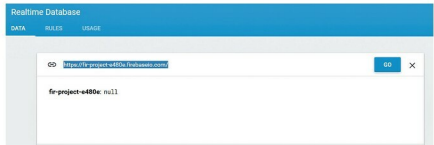


Figure 9: URL of the project

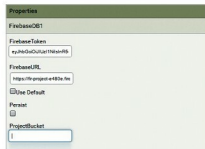


Figure 10: Set the URL to the Firebase component

editor. I hope you remember how to switch from the designer to the block editor. There is a button available right above the *Properties* pane to do this.

Block editor blocks

I have already prepared the blocks for you. All you need to do is drag the relevant blocks from the left side palette and drop them on the viewer. Arrange the blocks the same way you see them in the image. I will explain what each block does and how it is called.

A description of the block

- All the lines in the block are easy to understand.
- As soon as the *Add* button is clicked, it will check whether the text box is empty. If it is, the user will be

prompted to provide some value.

- If the value of the text box is not empty, then the name will be added to the student list and the same list will be saved into the phone database so that the next time the application is opened, the list will be the same.

Application code

- It was not possible to snap each and every block of the logic so I